

MIKKELI, Finland

WMO: 029470

Lat: 61.689N

Lon: 27.202E

Elev: 329

StdP: 14.52

Time Zone: 2.00 (EUE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.8	(c) -11.2	(d) -22.7	(e) 1.6	(f) -17.4	(g) -15.8	(h) 2.3	(i) -11.2	(j) 16.4	(k) 28.1	(l) 14.8	(m) 28.2	(n) 1.2	(o) 270	(p) 0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	80.6	65.7	77.2	63.7	74.0	62.1	68.3	76.7	66.1	73.9	64.0	71.3	6.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.8	72.0	63.0	87.0	69.7	61.0	81.1	67.4	32.7	76.6	31.0	74.0	29.4	71.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	13.2	11.7	-21.7	84.1	7.1	3.3	-26.8	86.5	-30.9	88.5	-34.9	90.3	-40.1	92.8
			-22.0	70.5	7.0	2.7	-27.0	72.4	-31.1	73.9	-35.1	75.4	-40.2	77.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	17.4	17.8	25.3	37.3	48.2	56.4	62.1	58.6	49.4	39.3	31.2	22.3		
	DBStd	17.85	13.46	12.98	9.47	6.91	7.61	6.40	5.66	5.50	6.41	7.48	8.38	12.48		
	HDD50	5057	1011	901	766	385	125	15	1	4	89	337	563	860		
	HDD65	9585	1476	1321	1231	831	521	267	126	208	469	797	1013	1325		
	CDD50	1002	0	0	0	4	70	207	375	271	70	5	0	0		
	CDD65	56	0	0	0	0	1	9	36	10	0	0	0	0		
	CDH74	708	0	0	0	0	42	136	399	131	0	0	0	0		
	CDH80	114	0	0	0	0	5	15	77	17	0	0	0	0		
(14) Wind	WSAvg	4.9	5.0	4.9	5.2	5.0	5.2	4.8	4.3	3.9	4.4	4.9	5.4	5.5		
(15) Precipitation	PrecAvg	24.4	1.9	1.5	1.3	1.3	1.6	2.6	3.1	2.7	2.0	2.4	2.2	1.9		
	PrecMax	30.7	3.4	3.1	2.5	2.5	4.6	5.3	5.5	5.4	4.5	5.9	4.5	3.4		
	PrecMin	19.4	0.3	0.2	0.1	0.5	0.4	0.7	1.3	0.7	0.4	0.5	0.4	0.6		
	PrecStd	2.8	0.8	0.7	0.6	0.6	0.9	1.0	1.2	1.3	1.1	1.2	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	39.6	50.2	66.9	78.9	82.0	86.0	81.9	70.2	57.3	48.6	43.9		
		MCWB	37.0	36.3	40.8	52.4	59.9	64.7	68.4	66.5	60.8	53.3	46.3	42.8		
	2%	DB	36.0	37.0	44.0	60.0	73.0	78.0	82.0	77.4	66.3	53.9	46.4	40.5		
		MCWB	35.2	35.5	37.5	48.4	57.0	62.6	66.9	64.4	58.7	51.3	44.0	39.6		
	5%	DB	34.6	35.0	40.3	54.9	68.3	74.2	78.5	74.1	63.2	52.1	44.3	38.1		
		MCWB	34.0	34.0	35.3	44.6	54.3	60.9	65.6	62.7	57.0	50.1	43.1	37.3		
	10%	DB	33.1	33.4	37.3	50.4	64.3	70.2	75.1	71.0	60.5	50.2	41.8	35.9		
		MCWB	32.6	32.7	33.9	42.4	52.4	58.8	63.8	61.6	55.8	48.6	41.0	35.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	37.5	42.1	54.5	62.6	68.6	71.3	69.7	62.2	54.3	47.3	43.1		
		MCDB	37.7	38.2	48.5	63.2	72.8	77.1	80.7	78.9	67.3	56.1	48.1	43.7		
	2%	WB	35.4	35.6	38.5	49.8	59.4	65.4	69.2	66.5	60.1	52.2	44.8	39.6		
		MCDB	35.8	36.6	42.6	57.9	69.3	73.4	78.3	74.0	64.6	53.7	45.6	40.2		
	5%	WB	34.0	34.1	36.3	46.0	56.4	62.7	67.4	64.4	58.3	50.7	42.8	37.3		
		MCDB	34.5	34.9	39.1	52.9	66.0	71.3	75.6	71.8	62.2	52.0	43.6	37.9		
	10%	WB	32.5	32.5	34.5	43.2	53.5	60.3	65.3	62.7	56.4	48.7	40.8	35.3		
		MCDB	32.9	33.2	36.6	49.3	62.6	68.5	72.8	69.0	59.8	49.9	41.7	35.9		
(35) Mean Daily Temperature Range	MDBR		10.7	12.9	16.5	18.1	21.1	19.3	19.1	19.3	16.4	10.3	7.6	9.5		
	5% DB	MCDBR	6.9	9.3	19.0	26.8	29.8	26.1	24.8	25.5	20.7	12.0	8.5	7.2		
		MCWBR	6.9	8.3	13.3	15.7	15.3	12.7	11.4	13.4	13.6	9.5	8.1	7.2		
	5% WB	MCDBR	7.4	8.2	14.8	24.2	25.8	22.0	21.7	21.2	18.0	9.8	8.1	6.7		
		MCWBR	7.5	7.6	10.9	14.9	14.3	11.9	11.0	12.1	12.5	8.4	7.9	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.293	0.320	0.360	0.356	0.353	0.377	0.362	0.338	0.319	0.269	0.358		
	taud		2.055	2.136	2.202	2.337	2.434	2.472	2.422	2.451	2.508	2.408	2.152	2.598		
	Ebn at Noon		140	209	249	263	274	277	266	261	247	205	131	91		
	Edh at Noon		12	22	30	32	32	31	32	29	23	17	11	7		
(44) All-Sky Solar Radiation	RadAvg		71	256	729	1175	1571	1663	1622	1248	719	296	79	36		
	RadStd		12	59	93	127	177	105	164	150	84	50	14	4		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+1.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	-295	N/A	N/A	N/A

Nomenclature: See separate page